

Yuchen Liu

 LinkedIn |  Github |  yuchen98@g.ucla.edu |  Personal Site

SKILLS SET

Programming Languages: Python (7 yoe), Golang (3 yoe), C/C++, Java, Bash

Infrastructure & Cloud: AWS (EC2, EKS, Lambda, CloudFormation), GCP, Kubernetes, Docker, Terraform, OpenShift, Jenkins (CI/CD pipelines), Linux (system-level debugging & automation), KVM

Distributed Systems & Dev Tools: Microservices, REST/gRPC APIs, Cloud SDKs, Observability Stack (Prometheus, Loki, Grafana), Service Reliability, Fault Tolerance, Load Balancing, Postman, Redis, MySQL, Neo4j

AI/ML Frameworks: PyTorch, TensorFlow, Relation Extraction, wandb, dspy (prompt engineering)

PROFESSIONAL EXPERIENCE

Meta Platforms, Inc

Software Engineer

01/05/2026 - Present

Menlo Park, CA

- Develop and optimize core modeling solutions for the Meta Recommendation System, focusing on scalable **Machine Learning** infrastructure and advanced algorithmic design.
- Collaborate with cross-functional teams to deliver high-impact **AI** products, leveraging expertise in **distributed systems** and machine learning technologies.
- Specialize in end-to-end model deployment, performance tuning, and reliability engineering for Meta's AI services.

MemVerge, Inc

Software Engineer

05/08/2023 - 01/03/2026

Milpitas, CA

- Designed and built a relation extraction pipeline from dialogs, injecting knowledge graph into **neo4j** and prompt engineering AI agents with **dspy** for improved query accuracy. [<link>](#)
- Led the design of a distributed automation framework for AI infrastructure products on **Kubernetes** and **EKS**, optimizing scalability for 10k **Rancher** clients.
- Developed custom **Kubernetes controllers** for AI infrastructure billing systems, integrating **MySQL** for automated data auditing.
- Engineered dynamic logging workflows with **Loki** and **Promtail**, enabling flexible log management.

Meta Platforms, Inc (Facebook) - Digital Commerce Platform

Software Engineer Intern

06/20/2022 - 09/02/2022

Menlo Park, CA

- Redesigned and developed payout settlement algorithm, reconciling daily transaction discrepancies of **\$80,000** and resolving tens of thousands of failures. Integrated **Python** and **MySQL** handlers into **C++** codebase.
- Developed algorithms to categorize transaction discrepancies across 3 platforms (Facebook, Google/Apple). Utilized **Python** for data processing and exported outcomes to relational databases.
- Automated data analysis to compare product ledgers with payment platform ledgers, identifying user fraud.
- Set up an alert service for immediate detection of financial inconsistencies.
- Wrote 2 **design documents** and conducted intern presentations.

EDUCATION

University of California, Los Angeles (UCLA)

Master of Science in Computer Science, GPA: **3.60/4**

03/23/2023

Los Angeles, CA

- Coursework: Software Engineering, Operating System, Machine Learning Algorithms, Large-scale Machine Learning, Data Mining, Quantum Programming, Advanced Computer Architecture, Algorithms

Dalian University of Technology

Bachelor of Electronic Information Engineering, GPA: **3.86/4**

06/01/2020

Dalian, China

- Coursework: Probability and Statistics, Complex Variable Function, Information Theory, Principles of Communication

COURSE PROJECTS

CS Master Capstone Project

CNN Compression Based on Collaborative Compression Method

Sept. 2022 - Mar. 2023

Los Angeles, CA

- Using **PyTorch**, pruned channels of the pre-trained **ResNet-56** to desired ratio in the guide of Collaborative Compression, fine-tuned the model for 100 epochs, loaded the model and evaluated on **CIFAR-10** data set.
- Analysed the compression ratios of 27 convolutional layers, pruned only every 3 layers, fine-tuned variations of compressed ResNet-56 and tested the differences of test accuracy after pruning each 3 layers.
- Discussed pros and cons of collaborative compression strategies and proposed various pruning approaches.